



## Distribution Network Model

### Problem overview

A distribution system has 7 DCs available and 9 stores that require the item under analysis. Table 1 provides the transportation/distribution costs per DC-store route. The fixed (weekly) and variable (per unit) costs per DC, as well as the available number of units are in Table 2, while the per store requirements (weekly demand) and penalties are presented in Table 3. A requirement is that either DC1 or DC6 must be operational. The company is planning its distribution network for the next three years. Demand may increase significantly and resources are available to expand up to two distribution centers. Management seeks a recommendation that balances cost, customer service, and network flexibility under future demand uncertainty.

Table 1. Transportation costs

| T   | s1      | s2      | s3      | s4      | s5      | s6      | s7      | s8      | s9      |
|-----|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| dc1 | \$ 10.5 | \$ 11.2 | \$ 11.7 | \$ 13.4 | \$ 12.9 | \$ 10.6 | \$ 7.4  | \$ 15.2 | \$ 7.3  |
| dc2 | \$ 7.1  | \$ 14.3 | \$ 15.4 | \$ 12.9 | \$ 14.1 | \$ 12.5 | \$ 10.1 | \$ 13.7 | \$ 14.6 |
| dc3 | \$ 7.3  | \$ 9.2  | \$ 7.3  | \$ 11.0 | \$ 7.4  | \$ 12.3 | \$ 12.8 | \$ 11.9 | \$ 12.1 |
| dc4 | \$ 15.7 | \$ 11.6 | \$ 13.9 | \$ 10.8 | \$ 11.2 | \$ 10.1 | \$ 9.1  | \$ 9.8  | \$ 8.2  |
| dc5 | \$ 12.9 | \$ 10.1 | \$ 9.8  | \$ 8.0  | \$ 7.3  | \$ 10.5 | \$ 7.4  | \$ 13.9 | \$ 13.8 |
| dc6 | \$ 10.6 | \$ 11.5 | \$ 8.0  | \$ 15.5 | \$ 7.3  | \$ 15.1 | \$ 15.9 | \$ 8.1  | \$ 15.3 |
| dc7 | \$ 14.4 | \$ 12.4 | \$ 15.6 | \$ 8.2  | \$ 15.0 | \$ 14.4 | \$ 10.3 | \$ 13.1 | \$ 9.3  |

Table 2. DC information

| DC  | Fixed cost | Variable cost | Available |
|-----|------------|---------------|-----------|
| dc1 | \$ 2,000   | \$ 12.0       | 450       |
| dc2 | \$ 3,300   | \$ 11.5       | 650       |
| dc3 | \$ 2,500   | \$ 15.4       | 660       |
| dc4 | \$ 2,900   | \$ 11.0       | 770       |
| dc5 | \$ 2,550   | \$ 12.0       | 660       |
| dc6 | \$ 1,960   | \$ 13.7       | 600       |
| dc7 | \$ 4,500   | \$ 11.9       | 870       |

Table 3. Units needed (demand) and penalty costs per units missing per store

| Store    | s1      | s2    | s3    | s4    | s5      | s6    | s7      | s8    | s9    |
|----------|---------|-------|-------|-------|---------|-------|---------|-------|-------|
| Penalty  | \$ 14.5 | \$ 18 | \$ 16 | \$ 15 | \$ 17.5 | \$ 19 | \$ 15.5 | \$ 17 | \$ 14 |
| Required | 560     | 780   | 550   | 590   | 240     | 870   | 100     | 190   | 150   |

**Demand Growth Cases.** The following four demand scenarios should be considered.

- No growth. Current demand levels.
- Uniform growth: 15% at all stores
- Urban growth: s2, s4, s8: +35%, all other stores remain unchanged.
- E-commerce shift: s1: +10%, s5, s7, s9: +50%, all other stores remain unchanged.

**Expansion of a DC:** When a DC is expanded its available capacity increases by 50% and fixed operating cost increases by 25%. For example, when considering the expansion of DC1, the available units would be 675 (instead of 450) and the fixed costs would be \$2,500 (instead of \$2,000).